Adheesh Mishra

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Experience

PharmaACE Pvt Ltd
Jun 2024 – Present

Senior Software Engineer *Oct 2025 – Present*

- Architected a configurable **salary increment platform** on **FastAPI** and **PostgreSQL**, replacing Excel-driven workflows with a rules engine covering eligibility criteria, hike slabs, approvals, effective-dated salary revisions, and audit history; cut policy rollout from ~2 months to under a day.
- Built asynchronous approval workflows on **Amazon SQS** with durable messaging, retries, idempotent consumers, and failure recovery, ensuring reliable salary revision processing and preventing duplicate approvals across distributed workflows.
- Implemented concurrency and consistency controls using **row-level locking**, **optimistic concurrency control**, and transactional guarantees to eliminate race conditions in salary revision and employee lifecycle flows; enabled point-in-time payroll reconstruction for audit and compliance.
- Replaced JSON-based payroll storage with normalized relational schemas and targeted indexing, tuning SQL execution plans to improve query latency by **35%** across **50K+ salary-history records for 3K+ employees**.
- Rebuilt the payroll analytics and reporting layer with server-side pagination and virtualized rendering, improving **p95 response time by 60%** on large result sets.
- Led an **8-member cross-functional engineering team**, owning architecture decisions, technical reviews, sprint planning, mentoring, and end-to-end delivery of core HR/payroll modules.

Software Engineer *Jun 2024 – Sep 2025*

- Engineered a reusable **faceted-query framework** powering **255+ FastAPI endpoints** across **11 analytics modules**, with dynamic filtering, tri-state select/exclude semantics, and server-side facet aggregation over **12 PostgreSQL schemas and 120+ tables**.
- Delivered Company, Product, and Disease Scanner modules across frontend and backend, covering **2,621 companies, 30K+ active pharmaceutical products** (including NDC-level generics) **and 5,035 diseases** through **70+ FastAPI endpoints**, with reusable filtering, comparison, and CSV/Excel export workflows spanning **20+ attributes**.
- Designed a hybrid pagination architecture over **300K+ clinical trial records**, pairing keyset pagination for sequential browsing with offset pagination for indexed page navigation; reduced deep-page query latency from **~1.8s to ~50ms**.
- Built analytics interfaces in **React** with **Redux actions/reducers/sagas**, **Chart.js**, and **React Virtuoso**, and authored a shared **React/TypeScript visualization library** of **25+ chart primitives** adopted across engineering teams, isolating API payload changes to a single normalization layer.
- Developed an LLM-powered **Retrieval-Augmented Generation (RAG)** pipeline on **Llama 3** with **OpenAI embeddings** and **FAISS** for semantic retrieval over proprietary pharmaceutical knowledge bases, surfacing relevant clinical context for analysts.

HealthAssure Pvt Ltd
Jun 2022 – Apr 2024

Software Engineer

- Owned the Distributor and Internal Manager portals end-to-end on **Node.js, React, PostgreSQL, and MongoDB**, serving **500+ distributors** and internal business teams.
- Shipped revenue features including rider add-ons, policy renewals, upgrades, lead-generation workflows, and agent dashboards, improving conversion and platform monetization.
- Optimized REST APIs, analytics dashboards, SQL queries, and database indexing strategies, reducing average API response times by **~45%**.
- Migrated production infrastructure from DigitalOcean and Heroku to **AWS** using **Terraform**, improving deployment consistency and operational reliability.

Skills

- **Languages:** Java, Python, TypeScript, JavaScript, SQL
- **Frameworks:** FastAPI, Node.js, Express.js, React.js, Redux, REST APIs, GraphQL
- **Databases:** PostgreSQL, MongoDB, Redis
- **Cloud:** AWS (EC2, ECS, Lambda, RDS, S3, API Gateway, IAM, SQS, SNS, CloudWatch), Docker, Nginx
- **Core CS:** System Design, Distributed Systems, Event-Driven Architecture, Concurrency Control, Database Design, Performance Optimization.

Education

B.Tech., SGSITS, Indore | CGPA: 7.2 / 10.00 *2018 – 2022*